MAX MELNIKAS

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Biostatistician & Machine Learning Engineer | EHR & Genomic Data | AI/LLM Development | Digital Health | Python & R

PROFESSIONAL EXPERIENCE

Mass General Brigham **August 2025 - Present**
Biostatistician *Boston, MA, USA*
- Architected an end-to-end PyTorch LLM fine-tuning and evaluation pipeline on high-performance computing GPU infrastructure, optimizing open-source foundational models for biological and clinical prediction tasks using large-scale metabolomics data
- Developed and validated novel DNA methylation risk scores utilizing the Illumina EPIC v1 array, successfully quantifying epigenetic risk profiles for asthma and multiple complex diseases.
- Conducted large-scale genomic association analyses on the RECOVER Pediatrics dataset, identifying key genetic variants and multi-omic correlates linked to Long COVID outcomes

Olira **June 2025 - August 2025**
Machine Learning Engineer Intern *New York, NY, USA*
- Engineered a FastAPI, compound-agentic-AI pipeline to enhance healthcare providers' EMR notes using patient data collected with a personal health assistant app and fetched from a MongoDB database
- Optimized model prompts to improve response accuracy while reducing token costs, leveraging a simulated LLM arena with custom Python-based and LLM-as-a-judge scorers through platforms like Braintrust and BigSpin
- Developed a front-end, Streamlit-based testing environment to visualize the performance of individual agent API calls

Harvard University **September 2023 - May 2025**
Data Scientist *Boston, MA, USA*
- Maintained an open-source, Python package coverting raw time series data collected through wearables into individual-level, meta-data summaries of physical activity, mobility, communications and survey responses
- Parallelized the computation of digital health summaries for 100+ participant clinical trials using Slurm to queue jobs on a high-performance computing cluster and presented results to multi-stakeholder research teams
- Developed a decision tree model to classify participant adherence to scheduled surveys, triggering alerts to study teams for nonadherent individuals, with 98% precision and 100% recall
- Collaborated cross-functionally with external research teams and internal software developers to triage and fix platform issues
- Stress-tested software updates to uncover potential bugs and audit Beiwe data quality during public releases of platform updates

Trinity Life Sciences **June 2024 - August 2024**
Summer Associate Consultant, Commercial Analytics *Waltham, MA, USA*
- Analyzed claims data of ~60k HCPs and identified 5,000 top promotional targets for a rare-oncology product launch
- Modeled several field team structures using a series of workload assumptions to generate a final sizing of 35 field reps
- Aligned the US into 35 territories rolled up into 5 regions, ensuring adequate workload and transportation access for reps
- Generated content and synthetic sales data for a demo of a pharmaceutical field team incentive compensation solution

EDUCATION

Harvard University **August 2023 - May 2025**
Master's, Biostatistics *GPA: 3.8*
- Relevant Awards: John E. Thayer Scholarship, Harvard Chan Central Grant Recipient

Brandeis University **August 2019 - May 2023**
Bachelor's, Neuroscience *GPA: 3.8*
- Relevant Awards: Magna Cum Laude, Dean’s List, Bauer Research Fellowship, Brandeis Presidential Scholarship, Guller Young Scholarship, Lithuanian Jewish Community Scholarship

SKILLS

Skills: Agentic-AI, Machine Learning, Deep Learning, Survival Analysis, Time Series, Bayesian Analysis, Interpretable AI
Programs: Python, R, Julia, MATLAB, Java, AWS, FastAPI, MongoDB, Github, Plink/Plink2, Slurm, Docker, Tableau
Soft-skills: Teamwork, Mentorship, Presentation, Client support, Slide development, Literature review, Documentation
Interests: Cooking, Baking, Basketball, Tennis, Biking, Poker, Chess, Traveling, Listening to podcasts and music